

STATE-WISE GST REVENUE ANALYSIS

Python Data Analytics Project

Data Cleaning • EDA • Statistical Analysis • Visualization
Growth Analysis • Seasonality • Anomaly Detection

1. Executive Summary

The State-wise GST Revenue Analysis in India Using Python project analyzes Goods and Services Tax (GST) collection data across Indian States and Union Territories. The project applies Python-based data Analytic techniques to clean, transform, explore, visualize, and statistically analyze GST revenue data.

The analysis covers temporal trends, regional revenue contribution, GST tax components, growth metrics, seasonal patterns, and anomaly detection. A Streamlit dashboard is also provided for interactive exploration.

2. Project Overview

The project transforms raw GST collection data into structured analytical outputs that can support business and policy-oriented decision-making. The analysis examines GST collection across different periods, States and Union Territories, and tax components including CGST, SGST, IGST, and CESS.

3. Data Source

Item	Details
Dataset Name	Tax Collection on GST Portal
Source	GitHub-hosted CSV dataset
Dataset URL	https://raw.githubusercontent.com/DurkadeviGanesanMCA/State-wise-GST-Revenue-Analysis-in-India-Using-Python/main/tax-collection-on-gst-portal.csv
File Type	CSV
Number of Records	3,570
Number of Attributes After Transformation	12
Geographical Coverage	Indian States and Union Territories, along with certain non-state entities present in the source data.

The datasets contains information about GST collections by date, state or territory, state code, and individual GST tax components.

4. Problem Statement

Government tax authorities collect large volumes of GST revenue data across different states and tax categories. Raw tax collection data alone does not provide sufficient insight into regional performance, revenue trends, seasonal patterns, or unusual fluctuations.

Without systematic analysis, it can be difficult to compare GST revenue performance across states, understand changes over time, identify the contribution of different GST components, detect unusually high or low revenue periods, measure YoY and MoM growth, understand seasonal patterns, and distinguish genuine revenue changes from incomplete or anomalous data.

This project addresses these challenges through Python-based data analysis, statistical methods, and visualization techniques.

5. Business and Analytical Objectives

- Compare GST collections across Indian States and Union Territories.
- Analyze GST collection trends by month and year.
- Identify the contribution of CGST, SGST, IGST, and CESS.
- Determine the highest and lowest revenue-contributing regions.
- Calculate Year-over-Year (YoY) growth in GST collections.
- Calculate Month-over-Month (MoM) growth in GST collections.
- Identify seasonal patterns in GST revenue.
- Detect unusually high or low GST collection periods.
- Generate actionable insights for revenue planning and policy decisions.
- Provide a foundation for future revenue forecasting and interactive reporting.

6. Dataset Attributes and Derived Features

Attribute	Type / Role	Description
GST Id	Identifier	Unique identifier for each GST collection record.
Date	Date	Date of GST collection used for time-based analysis.
State Name	Categorical	Name of the Indian State or Union Territory.
State Code	Categorical	Official state or Union Territory code available in the dataset.
CGST	Numeric	Central Goods and Services Tax collected.
SGST	Numeric	State Goods and Services Tax collected.
IGST	Numeric	Integrated Goods and Services Tax collected from interstate transactions.
CESS	Numeric	GST Compensation Cess collected on specified goods and services.
Total_GST	Calculated	CGST + SGST + IGST + CESS.
Year	Derived	Year extracted from the Date field.
Month	Derived	Month extracted from the Date field.
Region_type	Derived	Classifies records as State or Union Territory.
Month_Year	Derived	Combines year and month for chronological monthly trend analysis.
Month_Number	Derived	Numeric month extracted from the Date field for seasonal analysis.
YoY_Growth_%	Derived	Measures annual percentage change in Total GST collection.
MoM_Growth_%	Derived	Measures month-to-month percentage change in Total GST

		collection.
Anomaly	Derived	Flags unusually high or low monthly GST collections using the IQR method.

7. Tools and Technologies

Tool / Technology	Purpose
Python	Main programming language for data analysis.
Pandas	Data loading, cleaning, transformation, grouping, and aggregation.
NumPy	Numerical operations and calculations.
Matplotlib	Static data visualization.
Seaborn	Statistical data visualization.
Plotly	Interactive charts and visualizations.
Google Colab	Development and execution environment.
GitHub	Repository hosting and source code management.
CSV	Dataset storage format.
Markdown	Project documentation and README creation.
Streamlit	Interactive dashboard development.

7.1 Python Libraries Used

Library	Purpose
Pandas	Data loading, cleaning, transformation, grouping, aggregation, and analysis.
NumPy	Numerical calculations and mathematical operations.
Matplotlib	Static charts and graphs.
Seaborn	Statistical visualizations and distribution analysis.
Plotly Express	Interactive visualizations.
Plotly Graph Objects	Advanced interactive visualization capabilities.
Datetime	Date and time-related operations.
Warnings	Suppression of unnecessary warning messages during analysis.

8. Data Pre-Processing and Feature Engineering

8.1 Data Ingestion

The CSV dataset was loaded directly from the GitHub repository using Pandas and then inspected to understand its structure and contents.

8.2 Initial Data Exploration

- Displayed the first five records.

- Displayed the last five records.
- Checked dataset information.
- Examined column names.
- Checked the number of rows and columns.
- Verified data types.
- Checked for duplicate records.
- Checked for missing values.
- Generated descriptive statistics.

The initial exploratory analysis helped establish the overall health and structure of the dataset.

8.3 Duplicate Check

Duplicate records were checked using Pandas. The final dataset contained 0 duplicate records, indicating that no duplicate rows were identified during validation.

8.4 Missing Value Check

Missing values were checked for every column. The final transformed dataset contained complete records based on the reported DataFrame information, with 3,570 non-null records across all 12 columns.

8.5 Column Renaming

Original Column	New Column
id	GST Id
date	Date
state_name	State Name
state_code	State Code
cgst	CGST
sgst	SGST
igst	IGST
cess	CESS

8.6 Date Transformation

The Date column was converted into a date format using Pandas. Date information was used to create Year, Month, Month_Year, and Month_Number fields.

8.7 Total GST Feature Creation

A calculated feature named Total_GST was created using the following formula:

$$Total_GST = CGST + SGST + IGST + CESS$$

This feature represents the combined GST collection recorded for each dataset entry.

8.8 Year and Month Extraction

Year and Month were extracted from the Date field to support annual aggregation, monthly trend analysis, YoY growth, MoM growth, and seasonal analysis.

9. Data Transformation and Aggregation

9.1 State-wise GST Revenue

GST collections were grouped by State Name and Total_GST was summed to determine cumulative revenue contribution by region. This analysis was used to compare the revenue contribution of different States and Union Territories.

9.2 Tax Component Analysis

GST revenue was analyzed separately by CGST, SGST, IGST, and CESS to identify how different GST components contribute to overall collections across regions.

9.3 Top 10 Revenue-Contributing States

The top 10 regions were identified by aggregating Total_GST and selecting the highest contributors. The analysis highlighted Maharashtra, Karnataka, Gujarat, and Tamil Nadu among the leading contributors.

10. Exploratory Data Analysis

Metric	Result
Records	3,570
Columns	12
Unique State Name values	39
Unique Date values	95
Unique years	10
Unique months	12
Region_type categories	2
Duplicate records	0

EDA included DataFrame structure verification, data type verification, missing value validation, duplicate validation, numerical statistical summary, categorical summary statistics, and unique value analysis.

11. Statistical Analysis

11.1 Measures of Central Tendency

GST Component	Mean	Median	Mode
CGST	620.19	218.61	0.00
SGST	802.64	331.61	0.00
IGST	905.85	299.20	0.00
CESS	299.32	10.55	0.00
Total_GST	2628.00	1003.99	0.00

The mean is higher than the median for all major GST components. This indicates positively skewed distributions, where a relatively small number of high-value observations increase the average.

The mode of zero indicates that zero-valued observations are frequent in the dataset.

11.2 Skewness Analysis

The analysis found strong positive skewness across the GST components and Total_GST. The reported skewness values ranged from approximately IGST: 3.23 to CESS: 18.01. This indicates long right-hand tails, with many observations having relatively low collection values and a smaller number having significantly higher values.

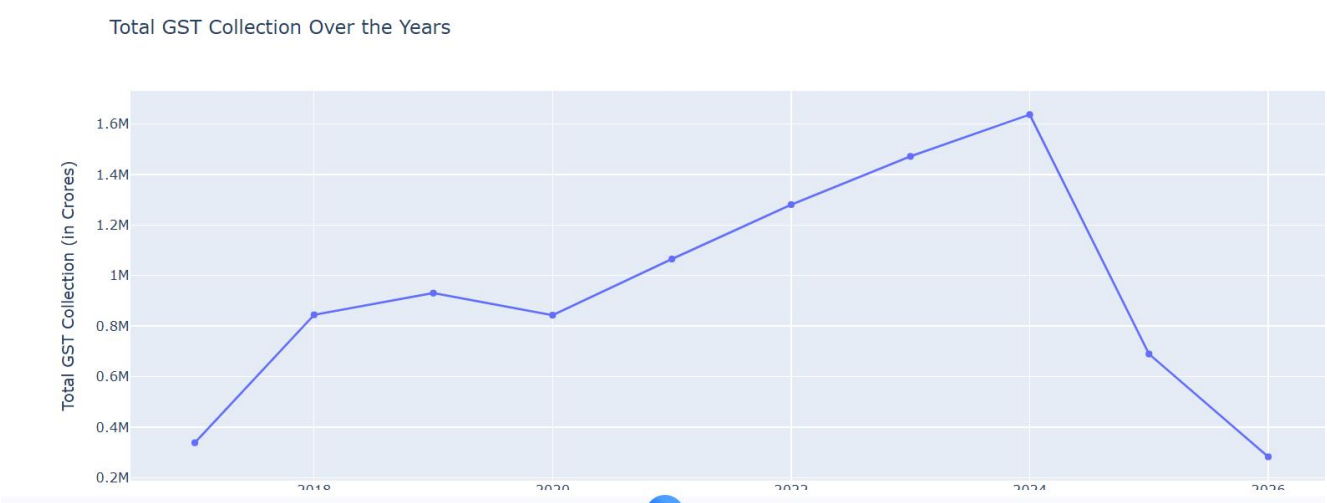
11.3 Kurtosis Analysis

Component	Kurtosis
CGST	19.30
SGST	15.56
IGST	14.20
CESS	465.88
Total_GST	29.07

The high kurtosis values indicate heavy-tailed distributions and the presence of extreme observations. CESS has particularly high kurtosis, suggesting exceptional values relative to the rest of the dataset.

12. Data Visualization and Analysis

12.1 Total GST Collection Over the Years

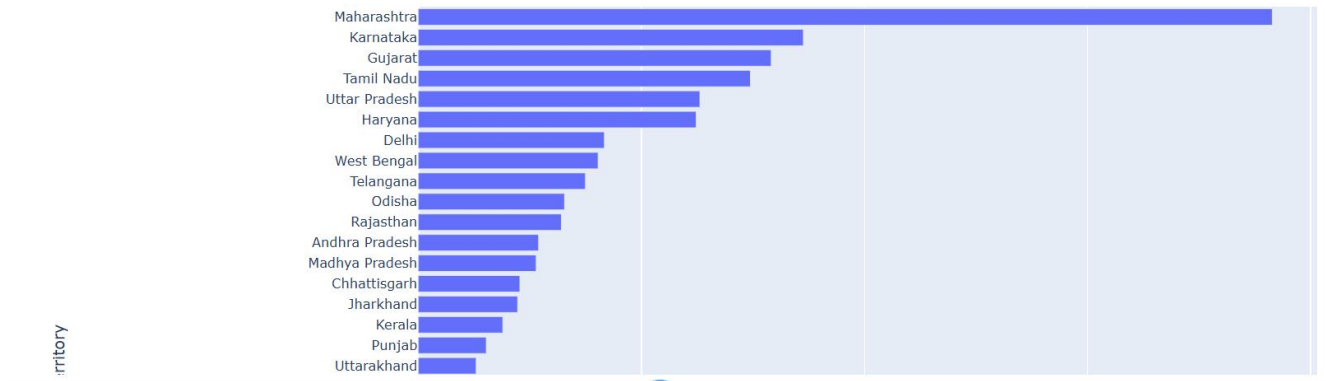


An interactive Plotly line chart was created to examine annual GST collection. The analysis indicates strong growth from 2017 to 2019, a decline or slowdown around 2020, significant recovery from 2021 onwards, continued growth through 2024, and sharp declines in 2025 and 2026.

12.2 Total GST Collection Across States and Union Territories

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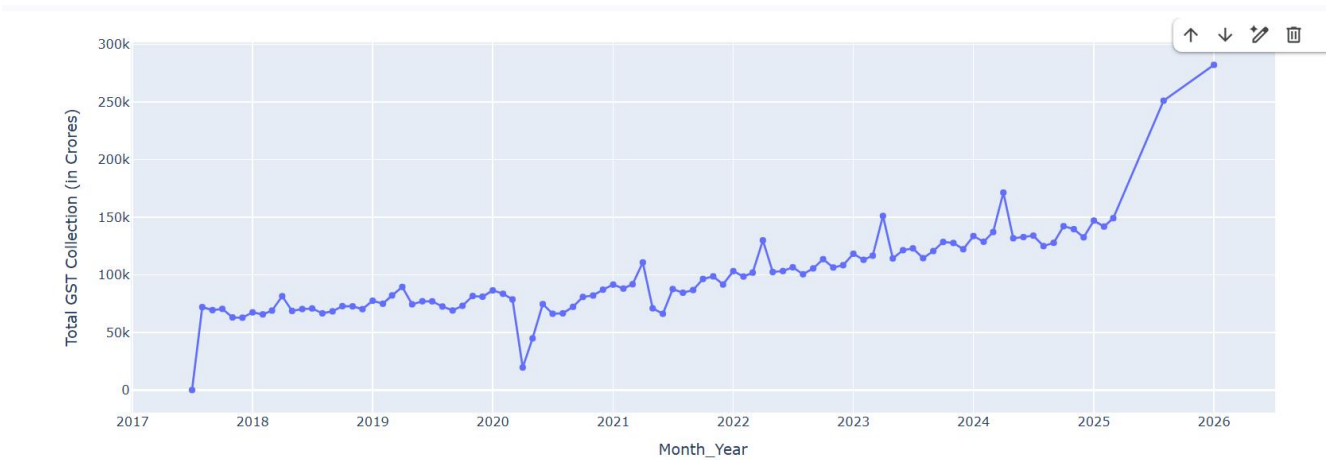
Total GST Collection Across States and Union Territories



The state-wise analysis identified significant differences in cumulative GST revenue. Leading contributors reported in the project include **Maharashtra (approximately 1,914,158.85 Cr)**, **Karnataka (approximately 862,921.76 Cr)**, **Gujarat (approximately 790,844.00 Cr)**, and **Tamil Nadu (approximately 744,217.29 Cr)**. Other significant contributors include **Uttar Pradesh, Haryana, Delhi, West Bengal, and Telangana**. Lower contributors include **Lakshadweep, Ladakh, Mizoram, and Andaman and Nicobar Islands**.

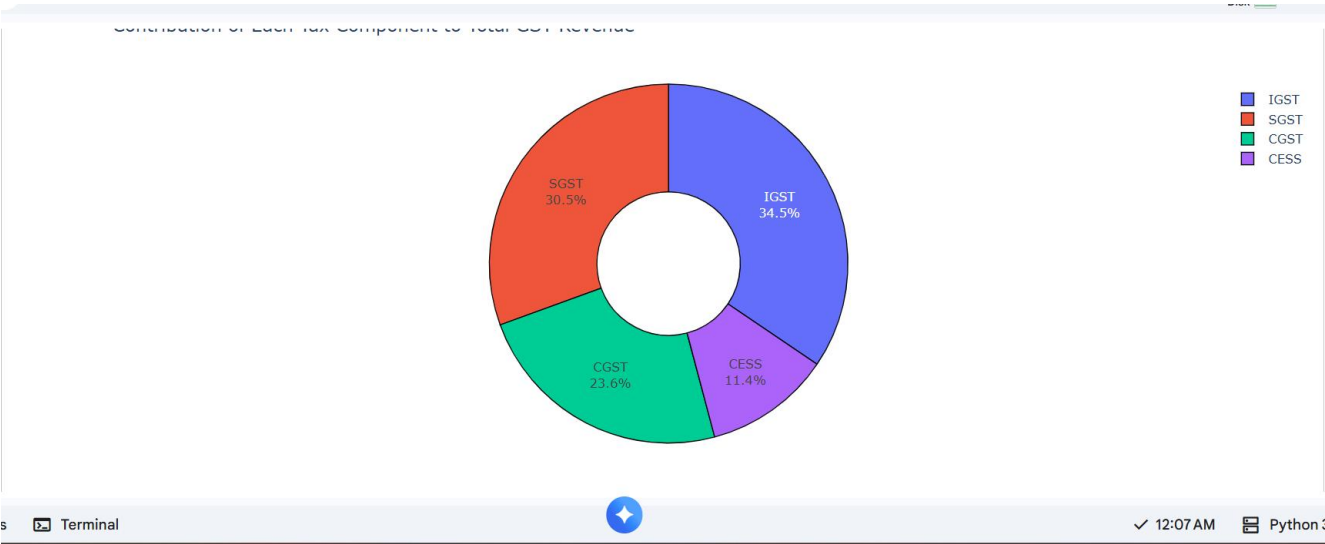
The results demonstrate substantial regional differences in GST revenue contribution.

12.3 Monthly GST Collection Trend



Monthly GST collections were aggregated using the Month_Year field to identify short-term revenue fluctuations, growth patterns, seasonal movements, high and low collection periods, and potential anomalies. The analysis indicates an overall upward trend with recurring fluctuations. However, later periods show significant declines that may be caused by incomplete data.

12.4 Contribution of CGST, SGST, IGST, and Cess to Total GST Revenue



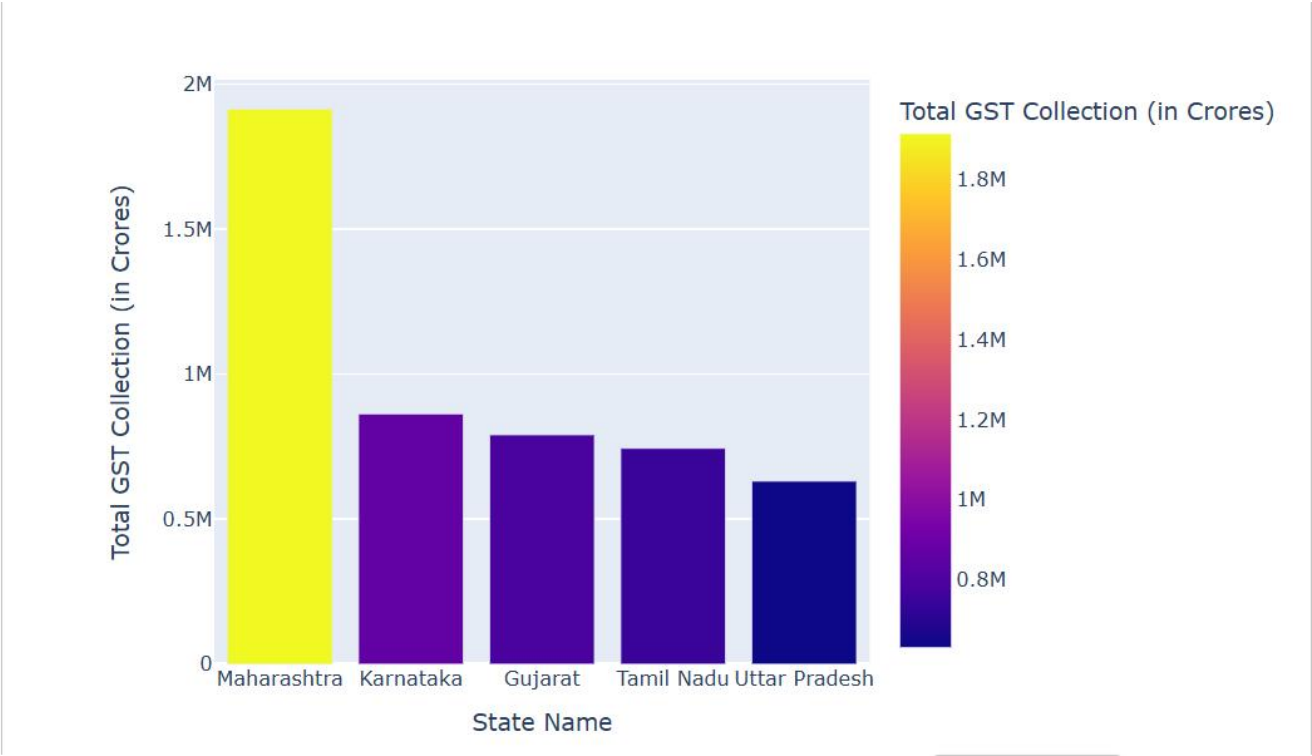
IGST (3,233,883.07 Crores): Integrated Goods and Services Tax typically accounts for the largest share. This is expected as IGST is levied on inter-state transactions and imports, and is designed to be a central clearing mechanism, often encompassing both central and state components initially.

SGST (2,865,428.56 Crores): State Goods and Services Tax represents the revenue collected by individual states from intra-state transactions. Its contribution is significant, reflecting the substantial intra-state economic activity.

CGST (2,214,063.99 Crores): Central Goods and Services Tax is the central government's share of intra-state transactions. Its contribution is generally comparable to SGST, as both are levied equally on intra-state supplies.

CESS (1,068,567.04 Crores): The Compensation Cess is levied on certain specified goods and services to compensate states for revenue losses due to GST implementation. While it contributes the smallest portion compared to the other three, its presence is crucial for state revenue stability.

12.5 Top 5 States by Total GST Collection

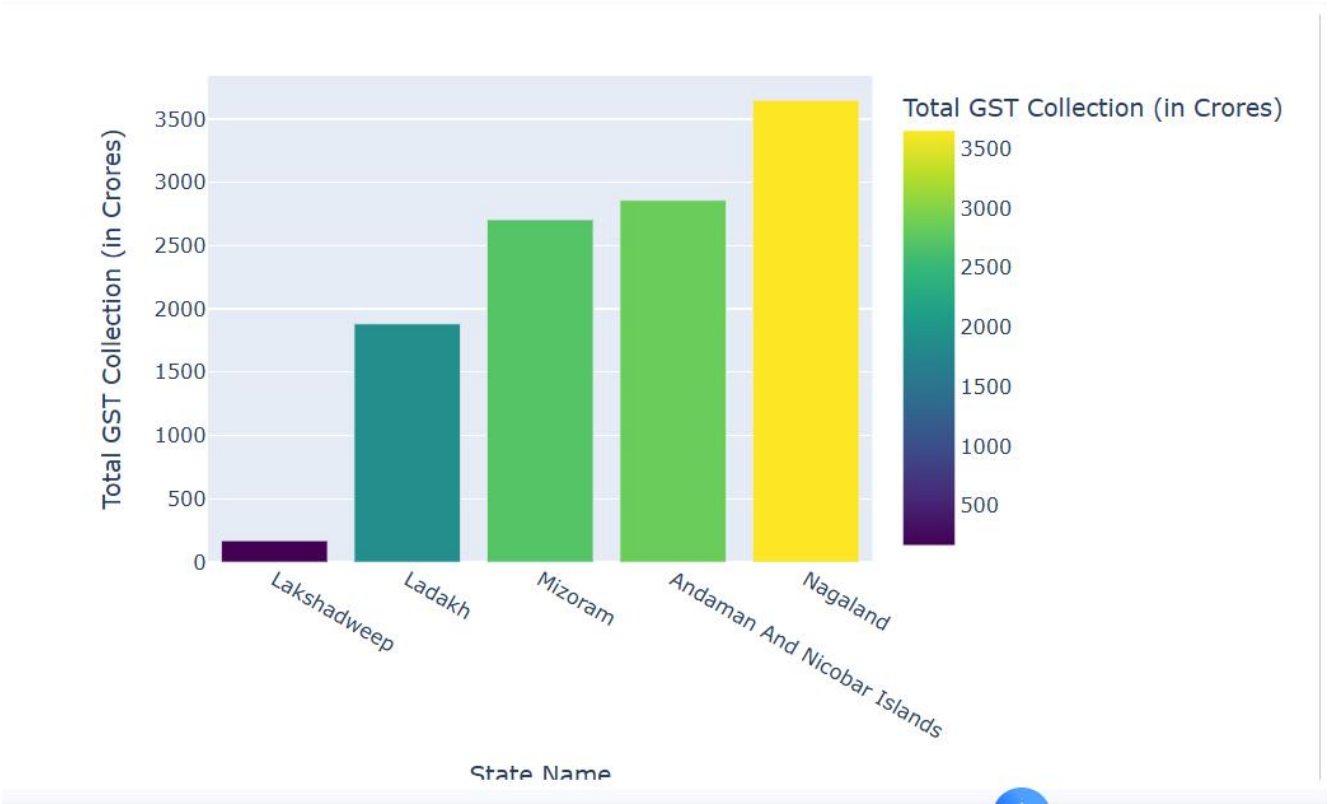


Maharashtra stands out as the highest contributor with **1,914,158.85 Crores**, underscoring its strong economic activity, industrial base, and large consumer market. This state consistently leads in various economic indicators in India.

Karnataka (862,921.76 Crores), Gujarat (790,844.00 Crores), and Tamil Nadu (744,217.29 Crores) follow closely, demonstrating robust GST collections. These states are known for their significant contributions to India's IT sector, manufacturing, and trade, respectively, which directly translate into higher tax revenues.

Uttar Pradesh (630,974.71 Crores) also features in the top 5, indicating its growing economic significance, large population, and increasing commercial activities.

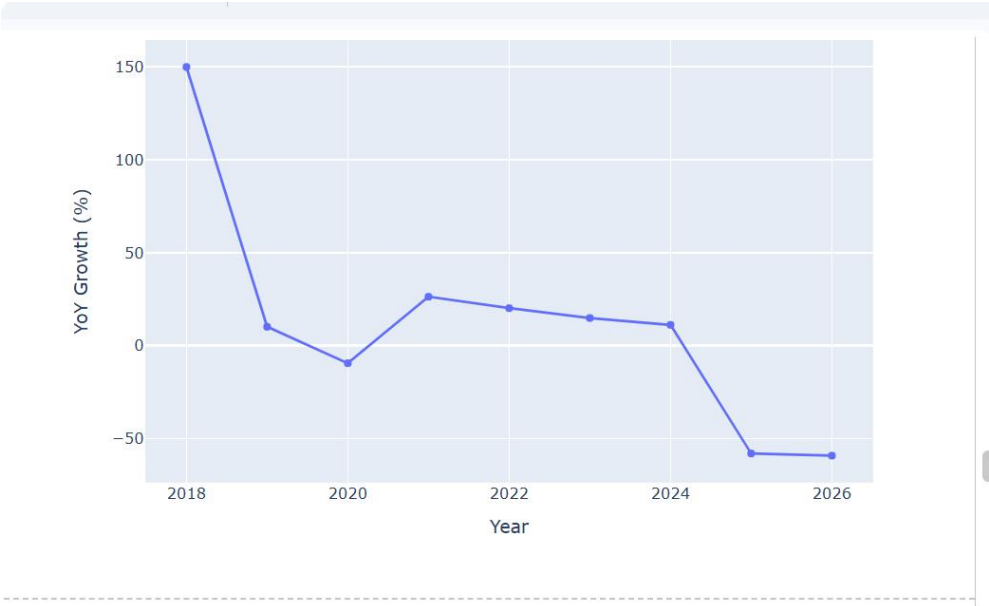
12.6 Bottom 5 States by Total GST Collection:



Lakshadweep records the absolute lowest GST collection with **170.03 Crores**, which is expected given its very small geographical area, population, and limited economic activity. Its revenue is significantly lower than other states/UTs.

Ladakh (1,882.48 Crores), Mizoram (2,704.45 Crores), Andaman And Nicobar Islands (2,858.07 Crores), and Nagaland (3,648.05 Crores) consistently rank among the lowest contributors. These are typically smaller **states** or **Union Territories**, often with challenging terrains, lower population densities, and economies that are not as industrialized or diverse as the leading states. Their economies might be heavily reliant on specific sectors like tourism or agriculture, with limited manufacturing or large-scale commercial activities.

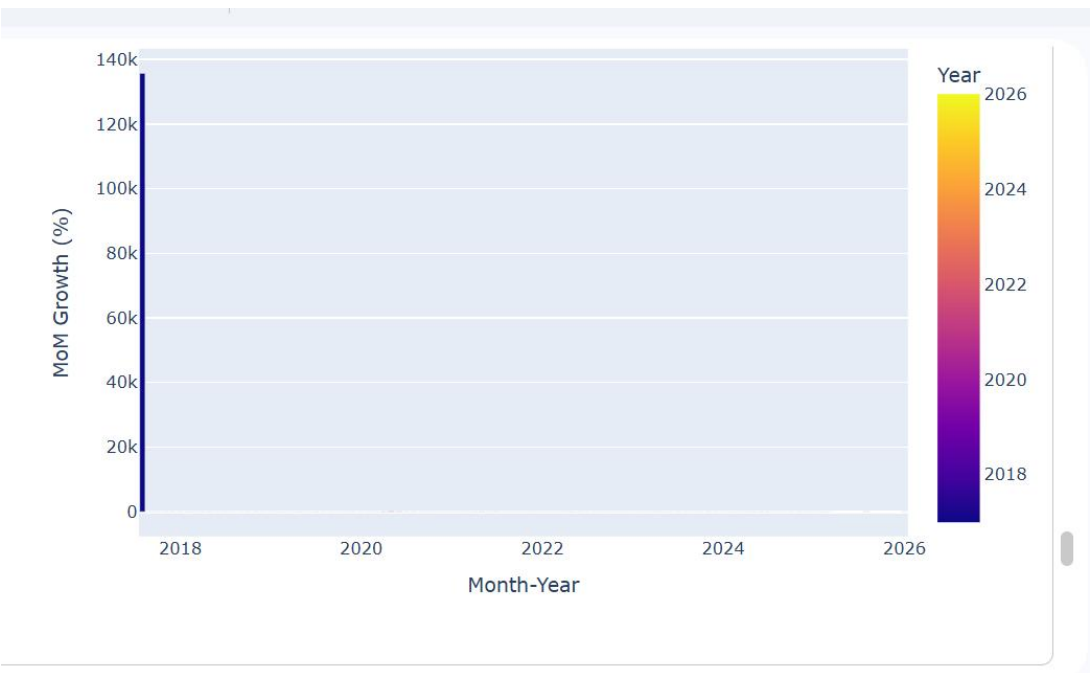
12.7 Year-over-Year Growth Analysis



Reported results include **2018: 149.96%, 2019: 10.18%, 2020: -9.32%, 2021: 26.35%, 2022: 20.22%, 2023: 14.91%, and 2024: 11.23%.**

The 2018 growth rate is influenced by the base effect because GST was introduced during **2017**. The negative growth in **2020** coincides with the period of **COVID-19**-related economic disruption. Strong positive growth after **2020** indicates recovery and continued expansion in the analyzed dataset. Reported declines in **2025** and **2026** should be treated cautiously because the project identifies possible data incompleteness.

12.8 Month-over-Month Growth Analysis



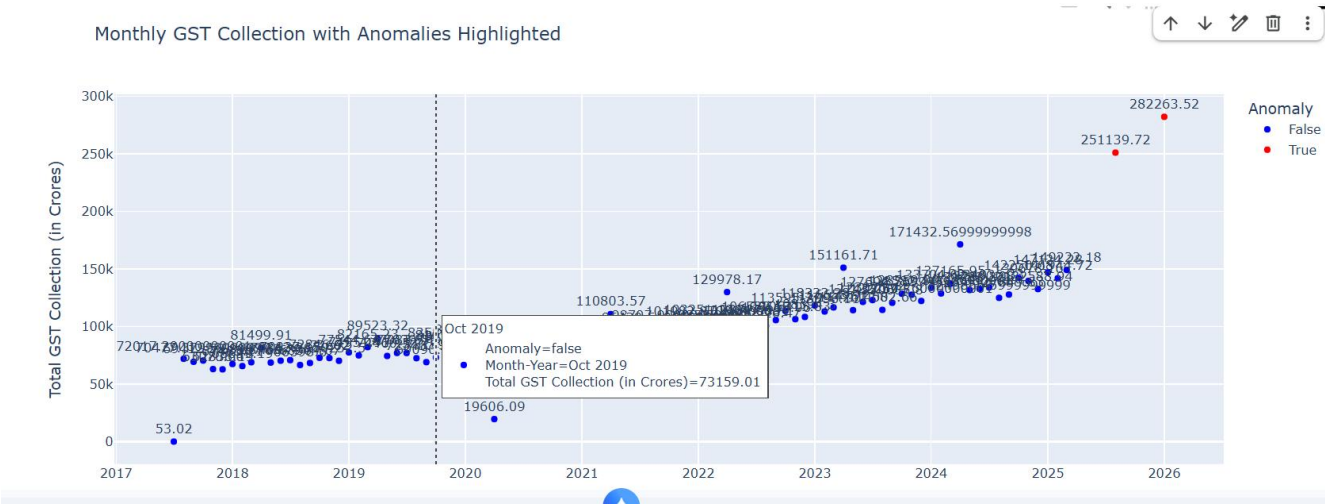
The chart reveals significant monthly volatility. For instance, the **MoM** growth has ranged from a low of approximately **-24.41%** to a high of **68.29%** (excluding the initial extreme growth due to base effect). This is common for tax collections, often influenced by payment cycles, seasonal consumer spending (**e.g., festival seasons, year-end sales**), and month-specific economic activities. Identifying recurring peaks and troughs helps in understanding inherent cyclical patterns.

12.9 Seasonal Analysis



Average GST collection was calculated for each month across the available years. Seasonal analysis helps identify recurring monthly patterns that may be associated with festival periods, fiscal year-end activity, consumer spending cycles, economic activity, and tax filing and payment cycles. Seasonality can be incorporated into future revenue forecasting models.

12.10 Anomaly Detection



The project used the Interquartile Range (IQR) method to identify unusual monthly GST collections. Observations outside these boundaries were flagged as anomalies. Anomalies may represent exceptional economic activity, large one-time collections, policy changes, compliance improvements, economic disruptions, or data quality issues. Each anomaly should be investigated individually before being interpreted as a genuine economic event.

13. Key Findings

- Overall GST Revenue Growth: GST collections show a general upward trajectory across the analyzed historical period, with a temporary decline around 2020 and a strong recovery afterward.
- Regional Revenue Concentration: GST revenue is highly concentrated among major economic states. Maharashtra is the leading contributor, followed by Karnataka, Gujarat, and Tamil Nadu in the reported cumulative analysis.
- Strongly Skewed Revenue Distribution: GST components and Total_GST show strong positive skewness, indicating many relatively low observations and a smaller number of very high-value observations.
- High Variability and Extreme Observations: High kurtosis values indicate heavy-tailed distributions and extreme observations, particularly for CESS.
- COVID-19 Impact: The reported YoY decline of -9.32% in 2020 coincides with the period of COVID-19-related economic disruption.
- Post-Pandemic Recovery: GST collections show positive YoY growth from 2021 onward in the reported analysis, indicating recovery and continued growth.
- Seasonal and Monthly Fluctuations: Monthly GST collections exhibit fluctuations that may be influenced by economic cycles, festivals, fiscal periods, and tax payment patterns.
- Data Completeness Is a Critical Concern: Sharp declines observed in 2025 and 2026 may be caused by incomplete data rather than genuine revenue deterioration. These periods require validation before forecasting or decision-making.

14. Analytical Framework

14.1 Descriptive Analysis — What Happened?

Descriptive analysis summarizes historical GST revenue patterns. The project examines total GST collection, state-wise revenue contribution, GST component contribution, annual revenue trends, monthly revenue trends, distribution of GST values, central tendency, skewness, and kurtosis.

14.2 Diagnostic Analysis — Why Did It Happen?

- COVID-19-related economic disruption and the 2020 revenue decline.
- Regional economic differences contributing to state-wise revenue disparities.
- Seasonal patterns affecting monthly collections.
- Potential data incompleteness causing recent-period declines.
- Extreme observations contributing to skewness and kurtosis.

These interpretations should be validated against external economic and policy data before being treated as causal conclusions.

14.3 Predictive Analysis — What May Happen?

The historical trend suggests potential continued GST revenue growth under stable economic conditions. Potential predictive applications include monthly GST revenue forecasting, annual GST revenue forecasting, seasonal forecasting, state-level revenue forecasting, and early detection of unusual revenue changes.

Reliable forecasting requires complete historical data and should account for economic shocks, policy changes, and data quality.

14.4 Prescriptive Analysis — What Should Be Done?

- Investigate unusually high and low GST collection periods.
- Monitor state-level revenue performance.
- Strengthen targeted compliance activities in areas with persistently low reported collections.
- Use seasonal patterns in revenue planning.
- Monitor YoY and MoM growth after major policy changes.
- Validate data completeness before forecasting.
- Investigate unusual records such as non-standard region names.
- Use interactive dashboards for continuous monitoring.

15. Recommendations and Decision Support

Improve Data Quality Controls

Before official revenue forecasting, validate data completeness, date coverage, state classification, duplicate records, non-standard entities, and partial-year data. This is particularly important for 2025–2026, where the project identifies potential incomplete data.

Focus on High-Contribution States

Revenue authorities should closely monitor high-contribution states such as Maharashtra, Karnataka, Gujarat, and Tamil Nadu. Understanding the drivers of their performance can support revenue planning, policy evaluation, compliance strategies, and replicable best practices.

Investigate Anomalies

High-value anomalies should be investigated to determine whether they resulted from one-time transactions, economic events, policy changes, or compliance initiatives. Low-value anomalies should be reviewed for economic slowdowns, data errors, reporting delays, or compliance issues.

16. Streamlit Dashboard

An interactive Streamlit dashboard was developed to support visualization and exploration of the GST revenue analysis.

Dashboard URL:

<https://state-wise-gst-revenue-analysis-in-india-using-python-ddfwyazy.streamlit.app/>

17. Project Workflow

Stage	Workflow Step
1	Data Source
2	Data Ingestion
3	Initial Data Exploration
4	Data Quality Validation
5	Data Cleaning
6	Date Transformation
7	Feature Engineering
8	Data Aggregation
9	Exploratory Data Analysis
10	Statistical Analysis
11	Visualization
12	Growth Analysis
13	Seasonal Analysis
14	Anomaly Detection
15	Business Insights
16	Recommendations
17	Interactive Streamlit Dashboard

18. Future Enhancements

- Add complete and regularly updated GST data.
- Validate and standardize all State and Union Territory names.
- Separate non-state entities from state-level reporting.
- Build automated data pipelines.
- Add advanced time-series forecasting.
- Apply machine learning models for revenue prediction.
- Develop state-specific forecasting models.
- Build a more comprehensive interactive dashboard.
- Automate monthly and annual reporting.
- Add external economic indicators for diagnostic analysis.
- Compare GST revenue with population, GDP, industrial activity, and other economic indicators.
- Implement automated anomaly alerts.
- Add role-based reporting for policymakers and analysts.

19. Conclusion

The State-wise GST Revenue Analysis in India Using Python project demonstrates how Python-based data analytics can transform GST collection data into meaningful insights. The analysis covers data ingestion, data quality checks, cleaning, transformation, feature engineering, exploratory data analysis, statistical analysis, visualization, growth analysis, seasonal analysis, and anomaly detection.

The findings show a broad historical upward trend in GST collections, significant differences in revenue contributions across regions, strong positive skewness in GST revenue distributions, and substantial variation in individual tax components.

The analysis also highlights the importance of data completeness, particularly when interpreting recent-year declines. The 2025–2026 results should be validated before being used for forecasting or policy decisions.

Overall, the project provides a practical analytical framework for understanding GST revenue patterns and supporting data-driven decision-making. With improved data validation, automated pipelines, advanced forecasting models, and richer external economic data, the project can be further developed into a comprehensive GST revenue intelligence and monitoring solution.